

**TEACHING MANUAL
ON
PRINCIPLES OF INTEGRATED PEST MANAGEMENT**



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PREFACE

The insects have assumed a special significance as pests in the context of changing cropping pattern. More than 40 per cent of world food production is lost to insect-pests, plant pathogens and weeds, despite the application of 2.3 million tons of pesticides to crops every year. The reasons for huge crop losses to pests despite the excessive use of pesticides may be due to pest resurgence, pest and disease resistance to pesticides, decline of biological control agents and other beneficial organisms. The theory and principles supporting IPM have been developed over the last ten decades. Contamination by pesticides chemicals in environment is a major global food safety issue, posing a serious threat to human health as well as other biodiversity. Protecting the ecosystem from the degrees of harmfulness of contaminated food chain with pesticides has become a daunting task and an environmental risk has emerged as a key problem for country in accordance to a number of studies. Degradation of environment due to pesticides is a worldwide problem and they have contaminated almost every component of our environment. Pesticides are poisons and can be hazardous. There is an urgent need of because Integrated Pest Management is an eco-friendly and sustainable approach to mitigate the pest incidence in crops. Integrated Pest management (IPM), which combines cultural, host resistance, biological, physical and chemical control strategies in a finest way rather than using a single component strategy proved to be more effective and sustainable. IPM is an ecologically and economically viable option to enhance the crop production and productivity, nutritional security, reduce the pest incidence and severity of the disease incited by the plant pathogen, upliftment of farmer community and reduce the environmental degradation and also play significant role as a sustainable tool in Indian context.

This manual has detailed information related to this course viz. pest, categories, causes for pest outbreak, losses caused by pests, pest surveillance and forecasting, tools or components of integrated pest management including physical, mechanical, cultural, biological, chemical, legal etc. The authors have tried to compile all these information systemically in a very simple language & comprehensive manner. We hope, this manual will prove helpful to the students and will update their knowledge.

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